

SmartFlow – A Database Driven Intelligent Traffic Management System

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Abstract:

Traffic congestion is a significant challenge in our modern society, causing increased travel time, fuel consumption, environmental pollution, and delayed emergency response. Conventional traffic management system often rely on fixed traffic signal timings and multiple cameras, which results in higher infrastructure cost and inefficient resource utilization. SmartFlow proposes a database driven intelligent traffic management system integrating AI, IOT and Database technologies for efficient traffic monitoring and intelligent decision support.

The system uses a single rotating IoT camera which monitors the 4 intersections. The camera rotates every 3 sec, capturing a 10-frame video clip from each direction. The clips are processed using YOLOv8 and OpenCV to detect and count vehicles, type of vehicles, ambulances and traffic congestion. The resulting information is transmitted through a FastAPI and stored in MongoDB Atlas, which contains real-time traffic records, historical and signal timings. A react based dashboard provides real time monitoring and historical analytics.

A custom SmartFlow Adaptive Priority Engine (SAPE) calculates the Traffic Priority Score (TPS) using:

$$TPS = (C \times 1) + (B \times 0.5) + (B_u \times W_{bu}) + (T \times 2.5) + (D \times 5)$$

The bus weight (W_{bu}) is 4 during rush hours and otherwise 2, while density (D) is scored from 1 to 3 where, 1 being the lowest and 3 the highest. There is an Emergency override here where once an Ambulance is detected, it is triggered. For normal conditions, green time is calculated using:

$$GreenTime(i) = MinTime + (TPS(i) / \sum TPS) \times (MaxTime - MinTime)$$

Where $MinTime = 20$ and $MaxTime = 60$. The system demonstrate how database management, AI and IoT can be used together for a scalable and cost effective smart traffic management system.